

PRASHANT MAURYA

AI Backend Engineer | Java, Spring Boot, AI-Integrated Systems

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PROFESSIONAL SUMMARY

Results-driven Backend Engineer with hands-on experience building production-grade microservices and AI-integrated systems using Java (Spring Boot) and Python. Proven track record designing REST APIs, JWT-secured architectures, semantic search pipelines, and agentic LLM workflows. Passionate about bridging backend engineering and applied AI to deliver reliable, scalable systems.

TECHNICAL SKILLS

Languages: Java, Python

Backend: Spring Boot, REST APIs, Microservices, MVC, Layered Architecture

AI / ML: LangChain, LangGraph, LLMs, RAG, Agentic AI(learning)

Security: Spring Security, JWT Authentication

Databases: MySQL

Tools & Other: Git, Postman, AZURE, Docker

EXPERIENCE

Web Developer — Technical Team, CHAD Foundation (NGO)

2026 – Present

HTML · CSS · Bootstrap · JavaScript · [chadindia.org](#)

- Spearheading development of CHAD Foundation's official website, building responsive, accessible web interfaces with HTML, CSS, Bootstrap, and JavaScript.
- Collaborating with stakeholders to deliver a scalable, production-ready platform showcasing the organization's initiatives and impact.

PROJECTS

Diagnyx — Agentic AI Medical Report Interpretation System | [Live](#) | [Github](#)

FastAPI · LangGraph · LangChain · Groq LLM · OCR · Pydantic

- Architected a 9-stage agentic AI pipeline (ingestion → reasoning → synthesis) for automated medical report analysis using LangGraph state machines.
- Engineered multi-step LLM workflow with iterative feedback loops, reducing hallucination rate through rule-based safety validation against clinical thresholds.
- Implemented multi-format document ingestion (PDF, DOCX, OCR images) with preprocessing pipeline; used structured Pydantic schemas for consistent API responses.

Lost & Found — AI-Powered Semantic Matching Microservices System | [Github](#)

Spring Boot · FastAPI · MySQL

- Designed dual-service architecture (Spring Boot + FastAPI) with REST inter-service communication, achieving clean separation of business logic and AI inference layers.
- Built semantic search engine using 384-dimensional sentence embeddings and cosine similarity; developed hybrid ranking algorithm (semantic + category + fuzzy location) achieving 85% match accuracy.
- Implemented JWT-based authentication with role-based access control (RBAC) and real-time matching pipeline with automated email notifications.

BloodConnect — Geospatial Blood Donation Platform Backend | [Live](#) | [Github](#)

Spring Boot · PostgreSQL

- Built RESTful backend with role-based access control (HOSPITAL / DONOR roles), enforcing blood compatibility rules and 56-day donation cooldown business logic.
- Implemented geospatial donor discovery using the Haversine formula for proximity-based matching; designed automated notification system for real-time donor alerts.
- Secured platform with JWT authentication and Spring Security.

EDUCATION

B.Tech — Computer Science & Engineering

2023 – 2027

Babu Banarasi Das University, Lucknow